

HAT-P-36b

R-1624

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_200414 · created 2026-08-02T04:55:33+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458953.9026 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.01
Transit depth (Rp/Rs)^2	2.33 +/- 0.32 [%]
Orbital Inclination (inc)	85.37 +/- 2.07
Airmass coefficient 1 (a1)	1.333 +/- 0.021
Airmass coefficient 2 (a2)	-0.1837 +/- 0.0076
Scatter in the residuals of the lightcurve fit is	1.54 %
Best Comparison Star	None
Optimal Aperture	5.98
Optimal Annulus	14.65
Transit Duration (day)	0.0946 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.01 vs 0.126 (deviation 2.7σ)
Duration fit vs published	0.0946 d vs 0.0925 d (deviation 0.44σ)
Mid-time O–C	6.2 min
Depth SNR	15.3
Residual scatter	1.54 %

Light curve

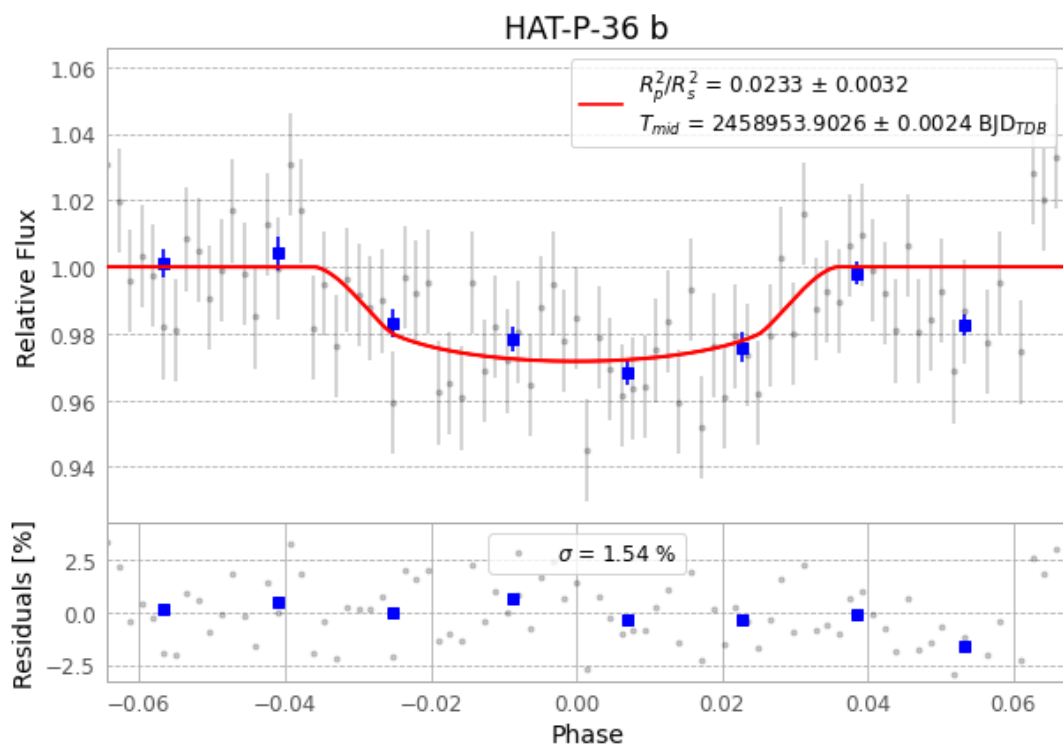


Figure 1 — FinalLightCurve_HAT-P-36 b_2020-04-14.png (SHA-256 e3d3691c34c9bfa1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [477, 286], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3081c58d346175ff...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), HATP-36200414073012.FITS → HATP-36200414114212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-200414.log (f60036375fb4...), FinalLightCurve_HAT-P-36 b_2020-04-14.png (e3d3691c34c9...), FinalLightCurve_HAT-P-36 b_2020-04-14.csv (82e726b2c288...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 04:55Z.